

Biochip Course

Electrochemical Detection

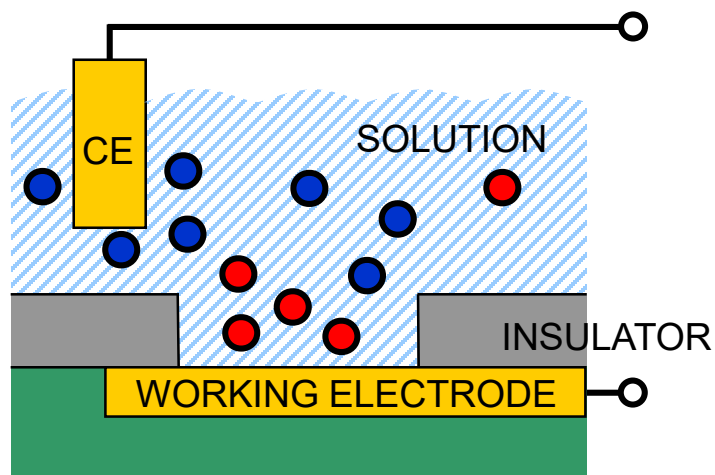
Introduction to the Electrochemical Interface

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a.y. 2022/2023

March 15st 2023

The Electrochemical Interface



In solution, charge is carried by ions that are heavier and bigger than electrons

$$\frac{m_{Cl^-}}{m_e} \approx 60000 !!!$$

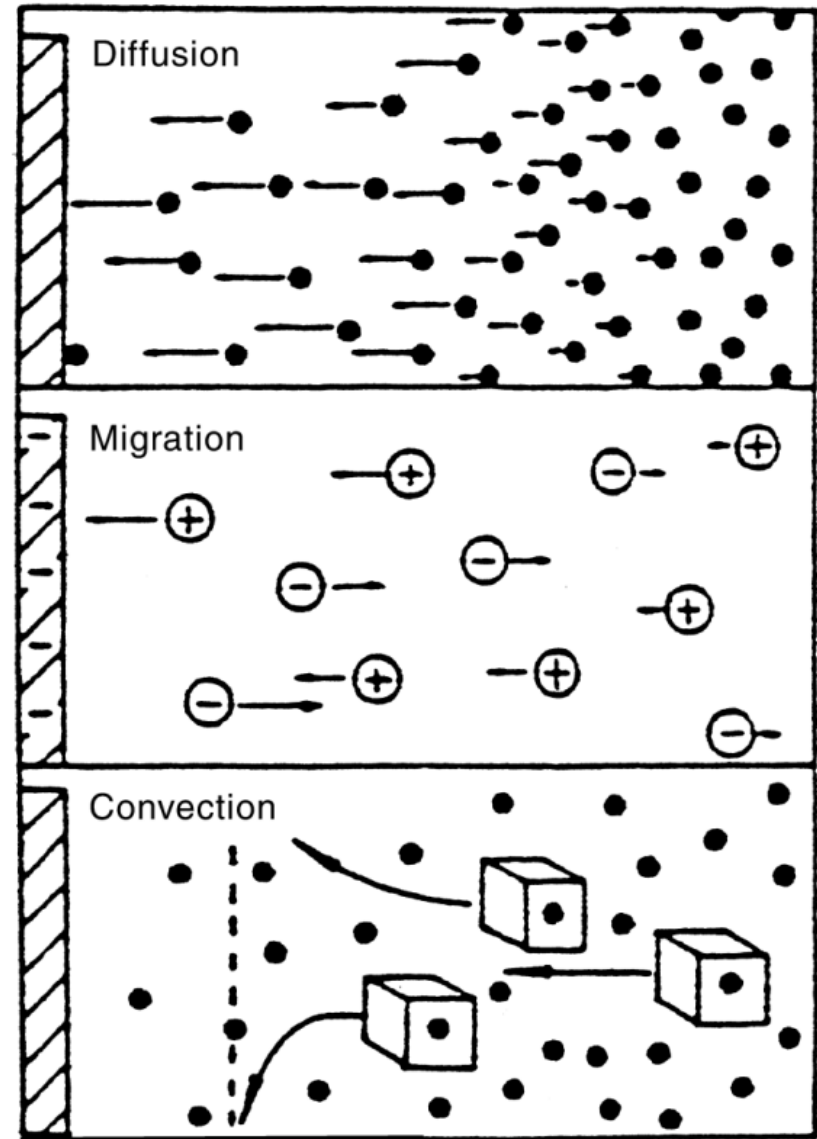
What happens to a **biased metal** in contact with an electrolyte:

- potential drop across the interface always
- redistribution of charge at the interface (*double layer*) always
- charge transfer (e^-) across the interface redox



Mass/Charge Transport Mechanisms

- Diffusion
- Migration (drift)
- Convection (fluid motion)
 - Natural (density gradient)
 - Mechanical (stirring, flow in microfluidic channel...)





Mass transport of **species i** described by Nernst-Planck equation:

$$J_i(x) = -D_i \frac{\partial C_i(x)}{\partial x} - \frac{z_i F}{RT} D_i C_i \frac{\partial \phi(x)}{\partial x} + C_i v(x) \quad [\text{mol/s/cm}^2]$$

C = molar concentration = mol/litre (or mol/ml = mol/cm³)

#molecules/cm³ = $C \cdot N_{Av} / 1000 = C \cdot 6 \cdot 10^{23} / 1000$

D = diffusion coefficient [cm²/s]

F = Faraday constant = $q N_{Av}$

R = gas constant = $N_{Av} k_B$

Z_i = # charge (dimensionless) of species i

$\Phi(x)$ = electrical potential

$v(x)$ = velocity of the solution

$$\left. \begin{array}{l} F = q N_{Av} \\ R = N_{Av} k_B \end{array} \right\} \Rightarrow F/(RT) = q/k_B T = 1/V_{th}$$



Charge Transport

By neglecting convection effects:

- no stirring / vibration
- stagnant layer near the interface (tens of μm for macroscopic setup)

$$\frac{I_i}{A} = -z_i F D_i \frac{\partial C_i(x)}{\partial x} + \frac{z_i F^2}{RT} D_i C_i E(x)$$

A = area

E(x) = electric field

Using Einstein relation ($D_i/\mu_i = k_B T/qz_i$, μ_i = mobility)

$$\frac{I_i}{A} = -z_i q D_i \frac{\partial p_i(x)}{\partial x} + z_i q \mu_i p_i E(x)$$

p_i = concentration
in #molecules/cm³

Same expression used in solid-state physics:

- many species i: $I = \sum_i I_i$
- charge carriers (ions) can have $|z_i| > 1$



Ion Mobilities

(low concentration, no inter-ionic interactions)

Ionic mobilities of various ions in water [19]

Cation	Mobility ($10^{-4} \text{ cm}^2/\text{Vs}$)	Anion	Mobility ($10^{-4} \text{ cm}^2/\text{Vs}$)
H^+	36.3	OH^-	20.5
Li^+	4	F^-	5.7
Na^+	5.2	Cl^-	7.9
K^+	7.6	Br^-	8.1
NH_4^+	7.6	I^-	8.0
Ca^{2+}	6.2	NO_3^-	7.4
Mg^{2+}	5.5	HCO_3^-	4.6
La^{3+}	7.2	SO_4^{2-}	8.3
Ag^+	6.4	$\text{Fe}(\text{CN})_6^{3-}$	10.5
$(\text{CH}_3)_4\text{N}^+$	4.7		

$$\mu \approx 5 \cdot 10^{-4} \frac{\text{cm}^2}{\text{Vs}}$$

$$D \approx 10^{-5} \frac{\text{cm}^2}{\text{s}}$$

Silicon:
 $\mu \approx 1000 \text{ cm}^2/\text{Vs}$
 $D \approx 20 \text{ cm}^2/\text{s}$



Example: Phosphate Buffered Saline

PBS: buffer solution commonly used for in-vitro biological research

Dulbecco's formula:

137mM NaCl; 8.10mM Na_2HPO_4 ; 2.68mM KCl;...

1 M means $N_A = 6 \cdot 10^{23}$ molecules per liter...

...concentration of ions $\approx 10^{20} \text{ cm}^{-3}$!



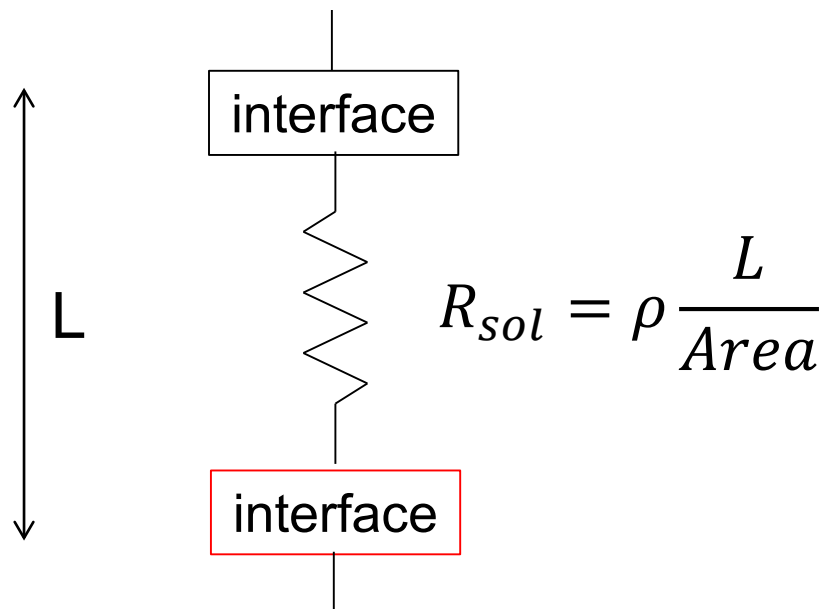
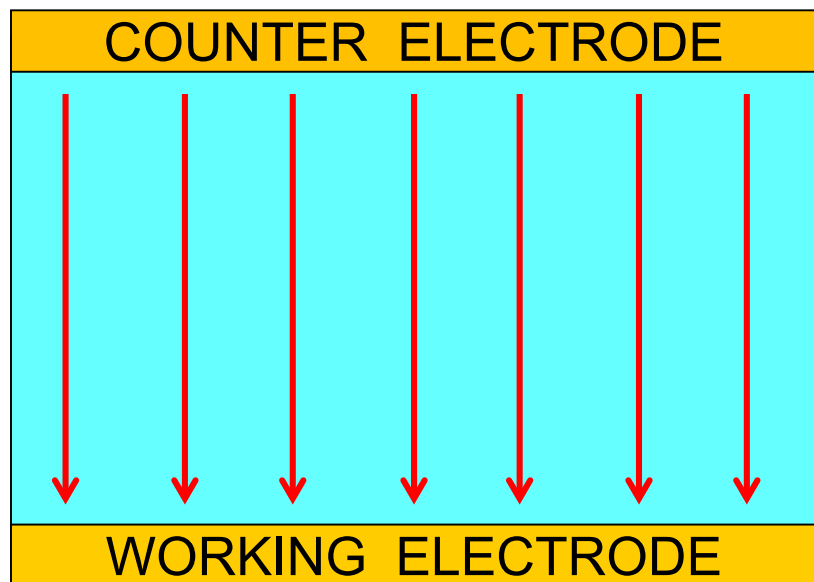
$\rho \approx 66 \text{ } \Omega \cdot \text{cm}$ (equivalent to silicon doped with $\approx 10^{14} \text{ cm}^{-3}$)

“good” conductor thanks to the high ion concentration



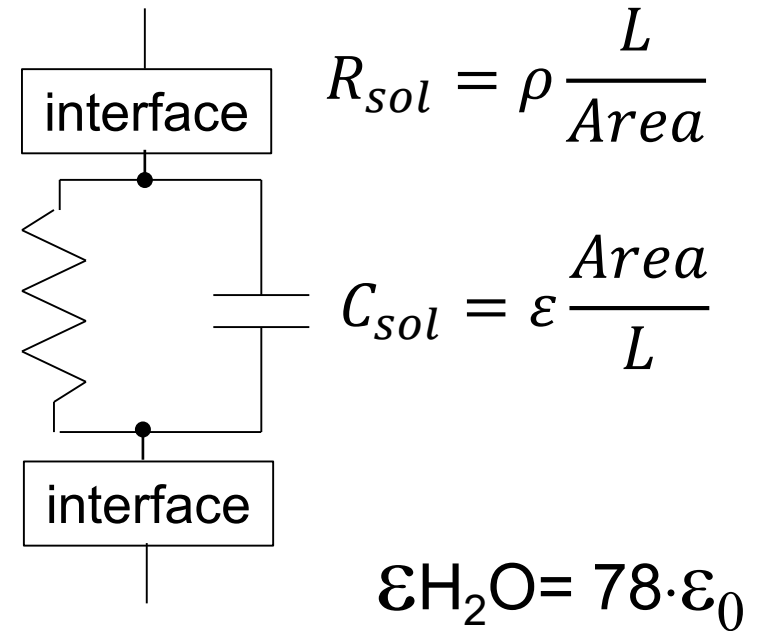
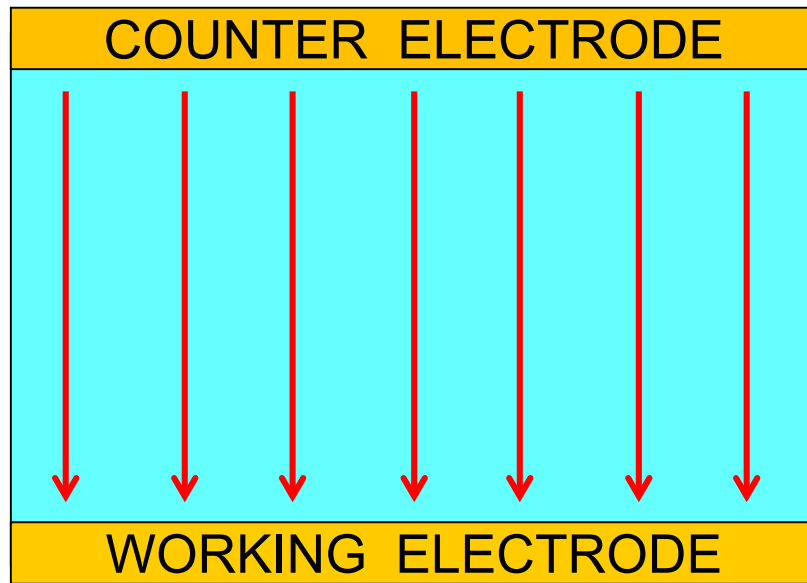
Equivalent Circuit of the Bulk Solution

For macroscopic electrodes, the whole solution volume contributes to the solution resistance:





Equivalent Circuit of the Bulk Solution



Dielectric relaxation time = $R_{sol} \cdot C_{sol} = \rho \cdot \epsilon = 0.5\text{ns}$



independent of geometry!

the bulk solution is a resistor up to $\approx 350\text{MHz}$ (for PBS)



Interface: Non-Faradaic Processes



Metal-Liquid Interface: the Double Layer

Metal

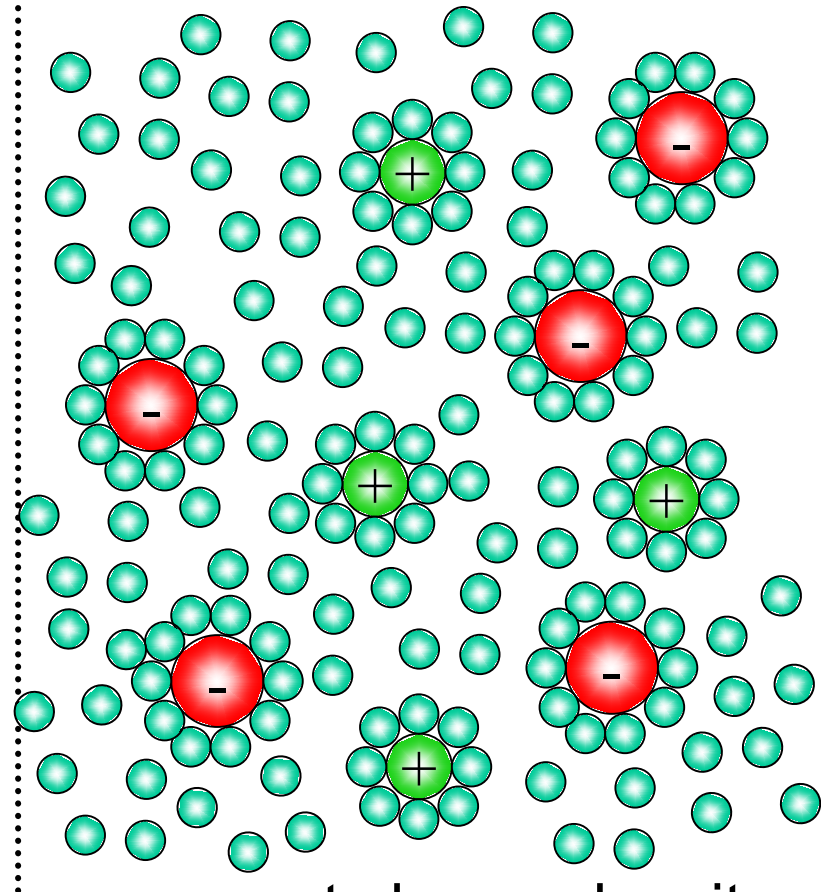
Interphase
region

anisotropic
forces on
electrons, ions,
water molecules

Free electrons
in a crystal

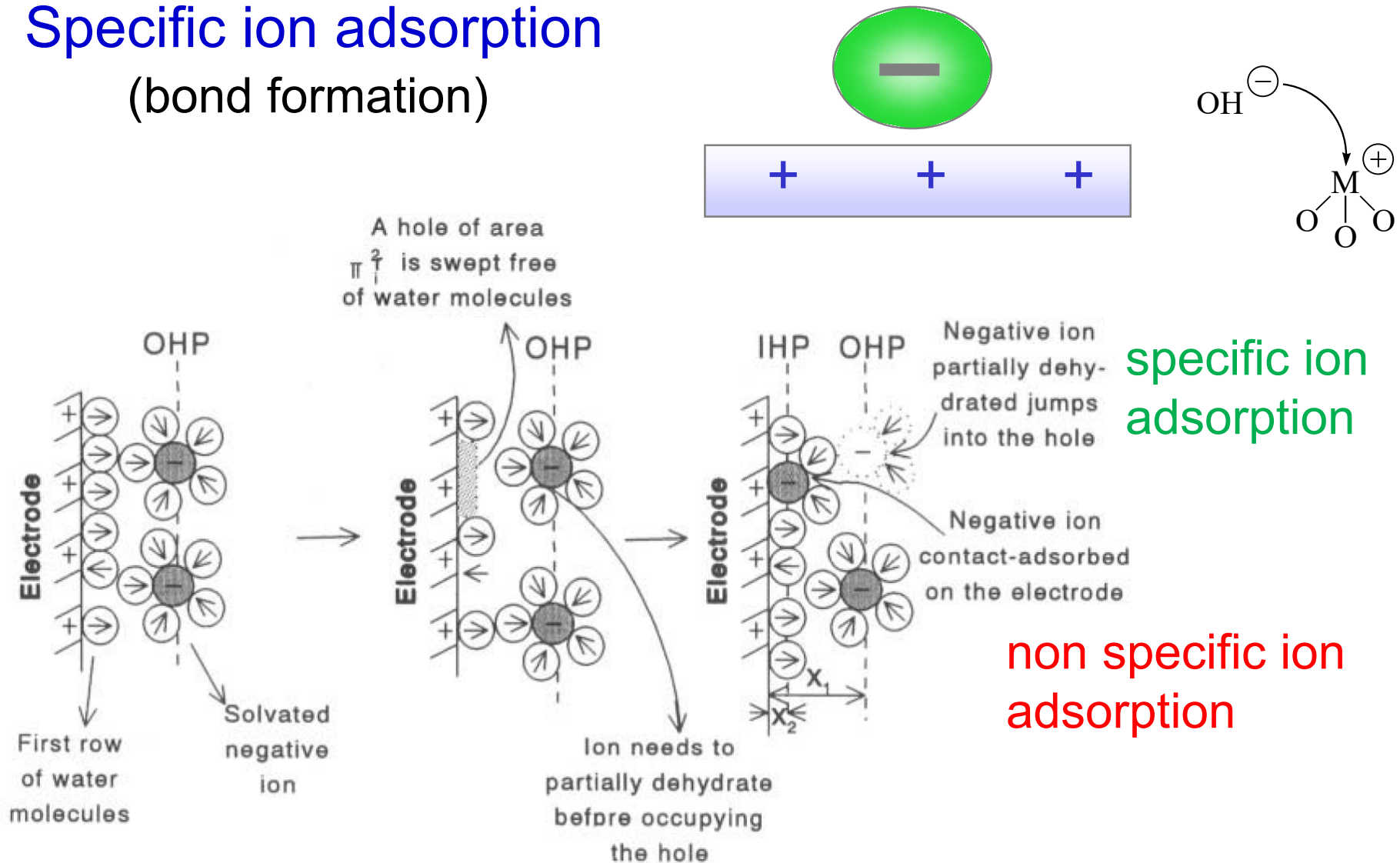
Bulk Solution

water
molecule



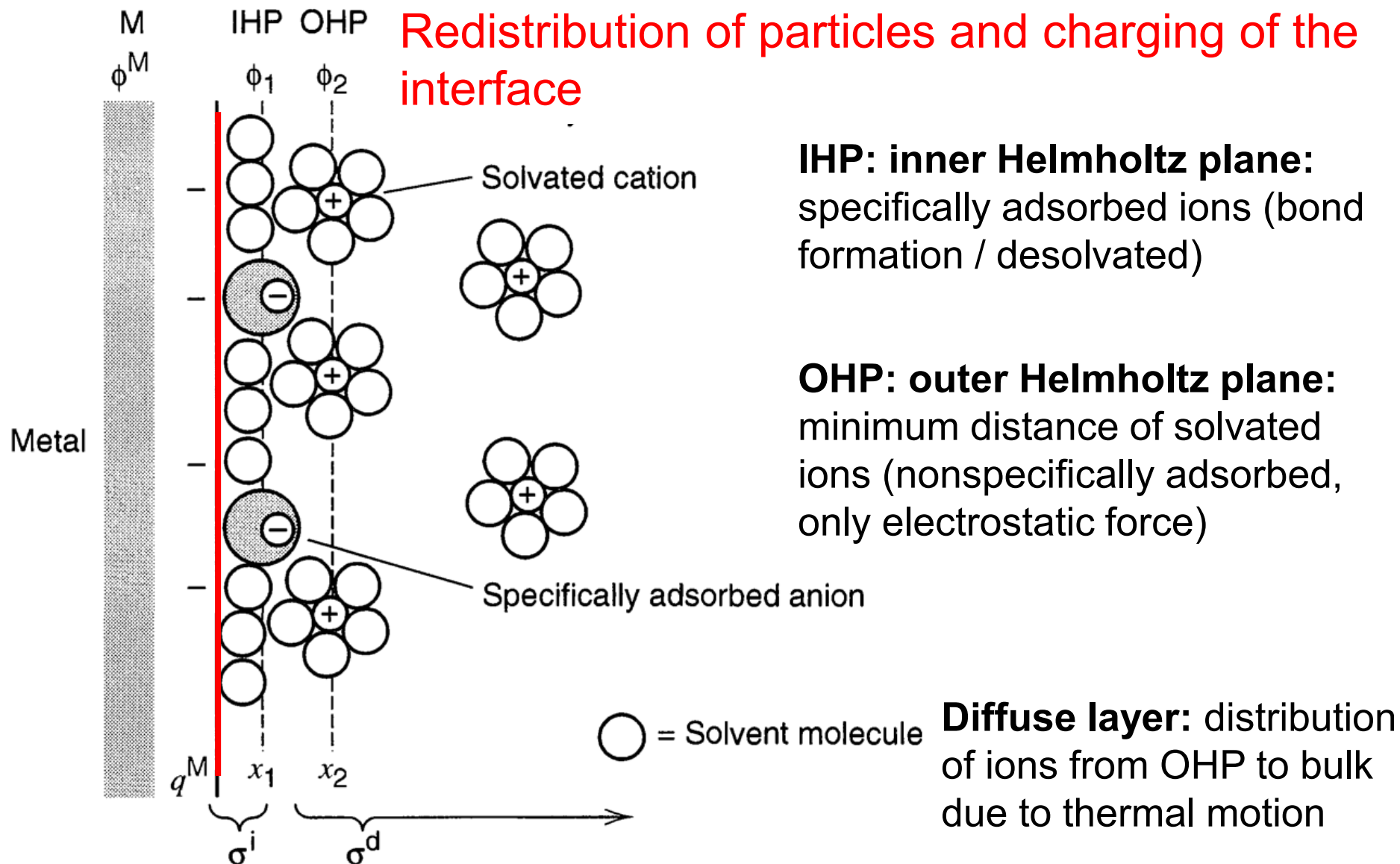
no net charge density
no net dipole orientation

Specific ion adsorption (bond formation)





The Double-Layer Structure

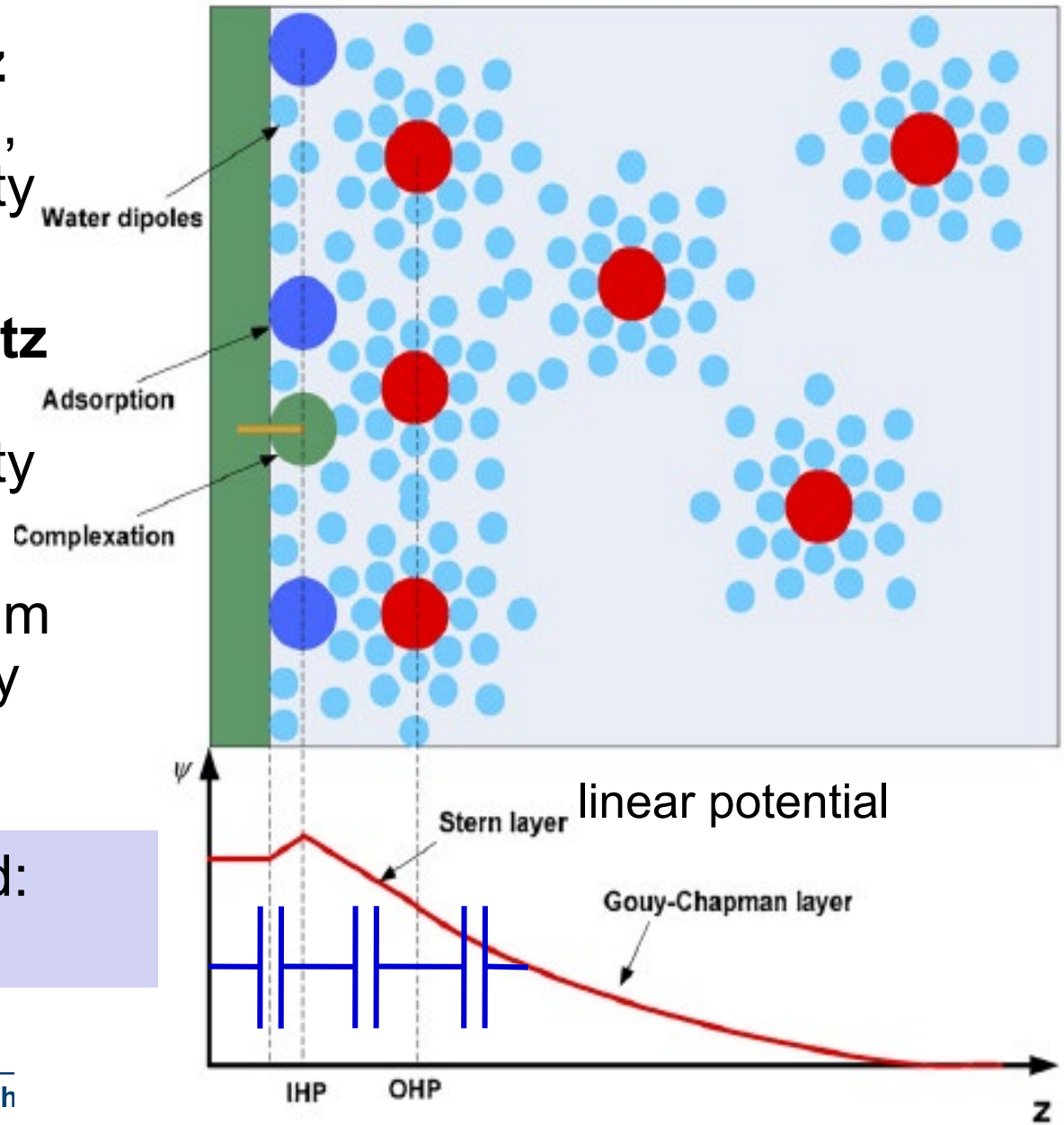




The Grahame Model

- **IHP - Inner Helmholtz Plane:** $\approx 0.1 - 0.2$ nm, constant charge density
- **OHP - Outer Helmholtz Plane:** $\approx 0.2 - 0.4$ nm, constant charge density
- **Diffuse layer:** $\approx 1-10$ nm, variable charge density

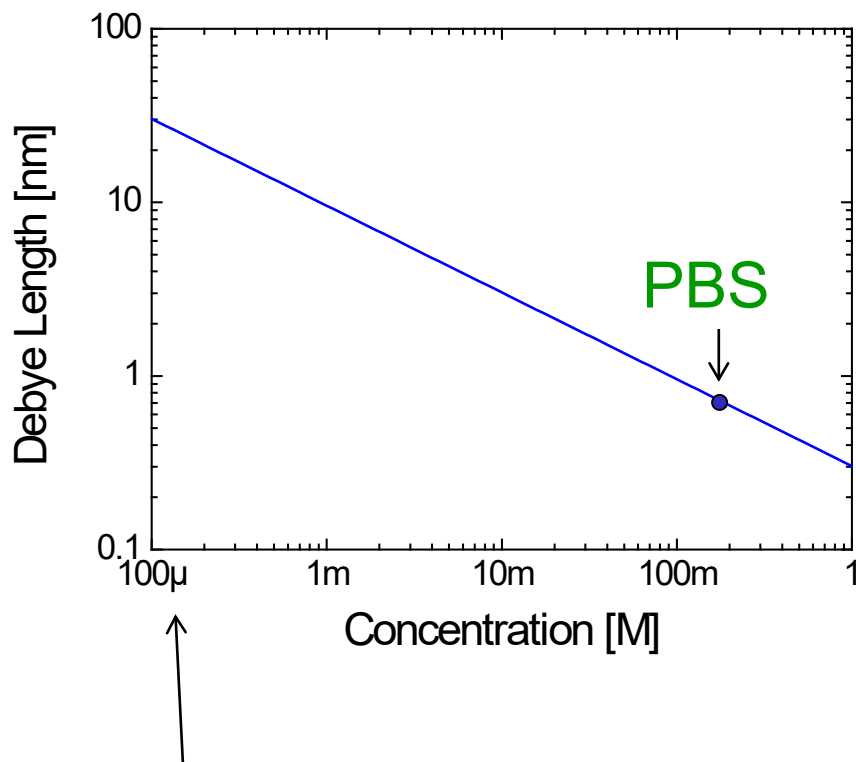
Interface electric field:
10-100 MV/cm !!!





Diffuse Layer

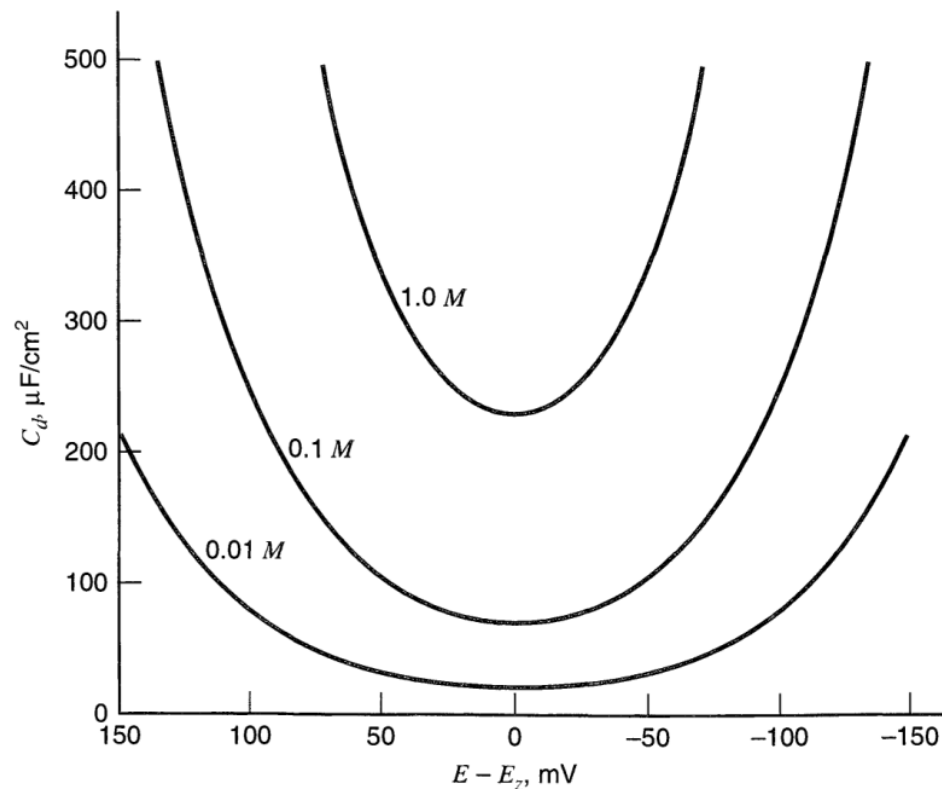
Debye length $L_D = \sqrt{\frac{\epsilon D}{\sigma}}$



i.e. 6mg/litre of NaCl

pure water, pH=7 (100nM): $L_D \approx 1 \mu\text{m}$

Capacitance

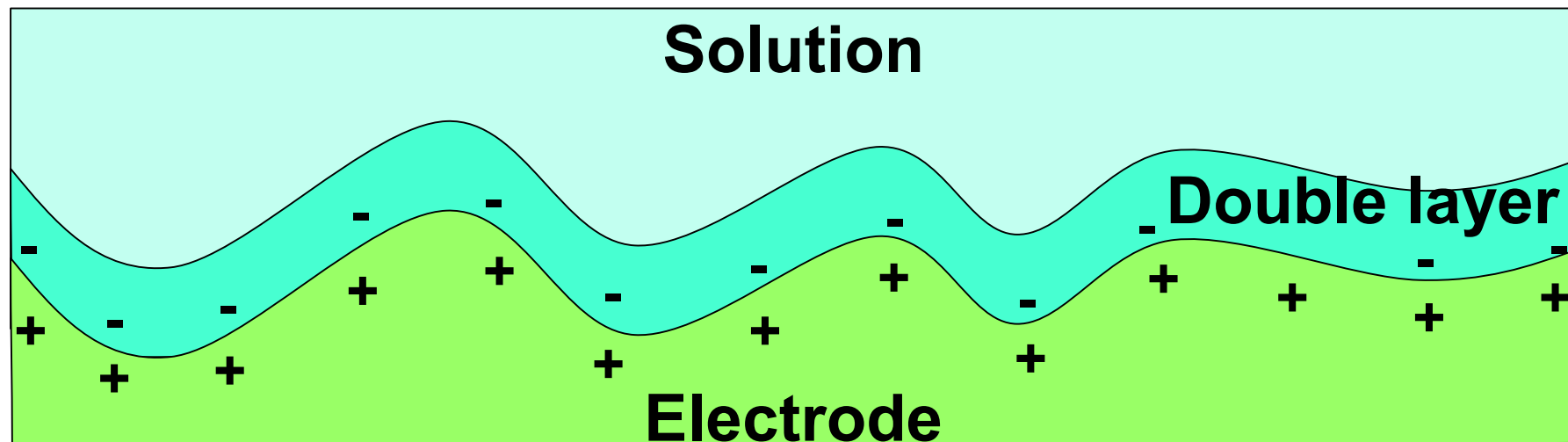


$$C_{dl} = 10 - 40 \mu\text{F}/\text{cm}^2$$
$$= 0.1 - 0.4 \text{pF}/\mu\text{m}^2$$



«Real Area» of an Electrode

C_{dl} depends on the first few nm of the interface



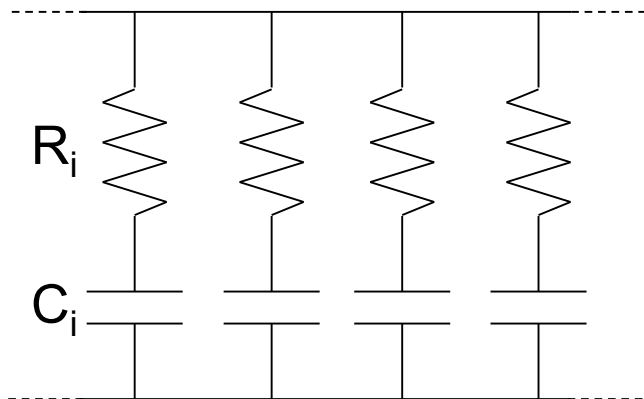
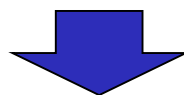
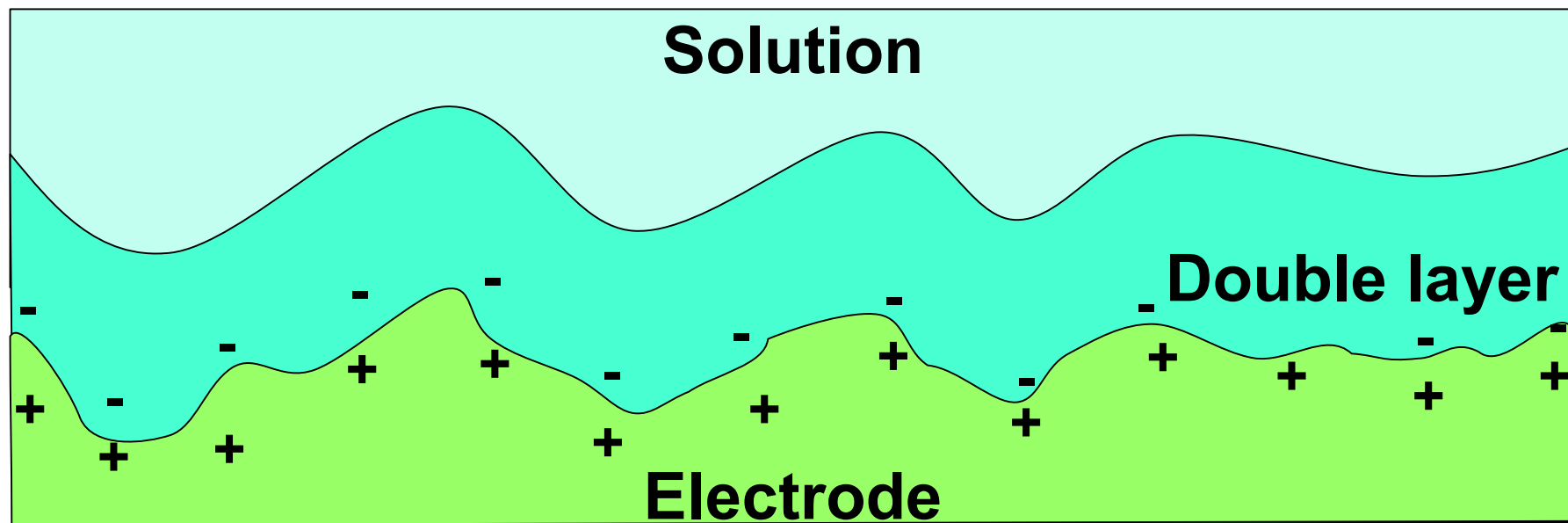
Double layer follows hills and valleys $\gg$ Debye length



atomic scale area for C_{dl} could be 2-3 times the “geometrical area”
depending on the **surface roughness**



Atomic Scale Disorder

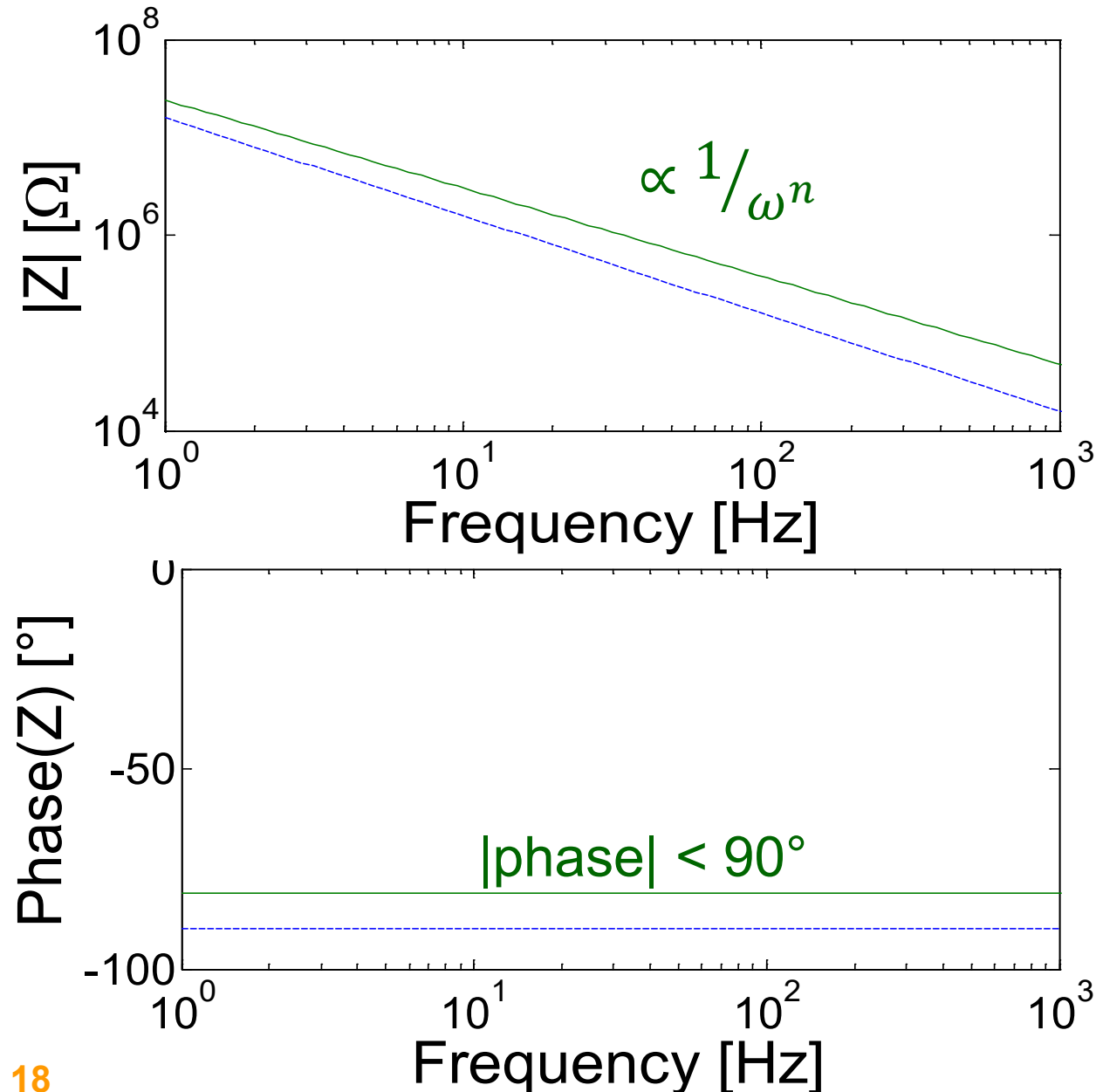


distribution of time constants

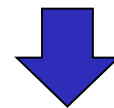
$$\frac{1}{Z} = Y = \sum_i \left(R_i + \frac{1}{sC_i} \right)^{-1}$$



Pseudo Capacitor: Constant Phase Element



“Slope” of C_{dl} is less than 1
($n = 0.8-0.9$)



$$Z_{CPE} = \frac{1}{Q(j\omega)^n}$$

surface disorder,
porous electrodes,
adsorption, ...

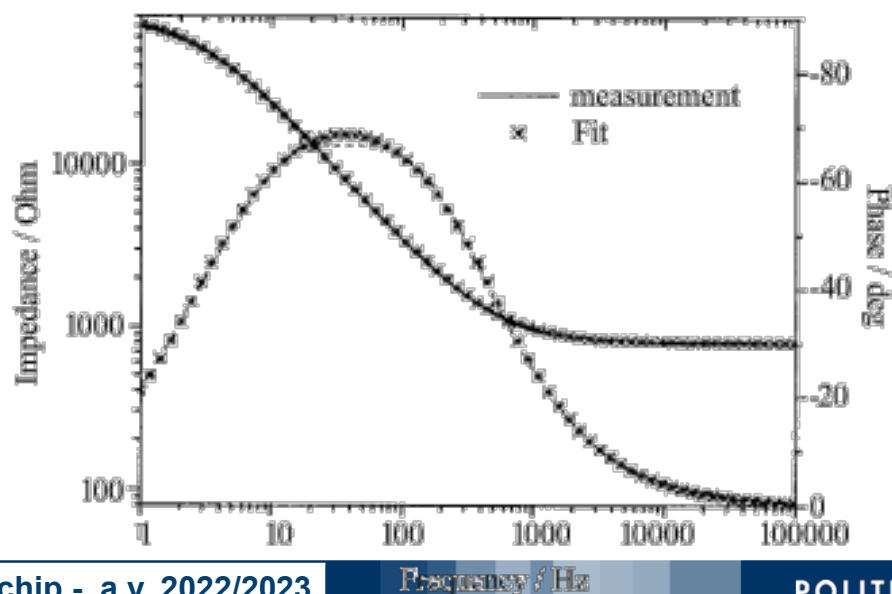
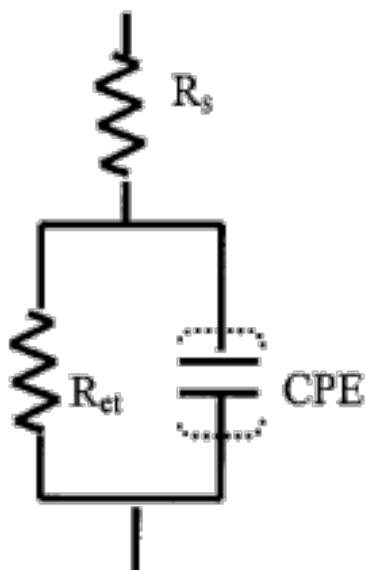
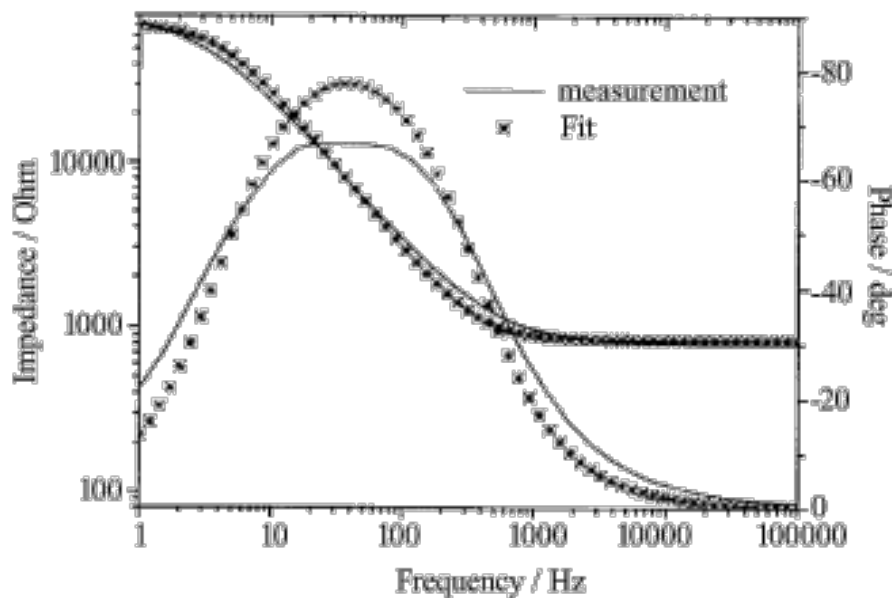
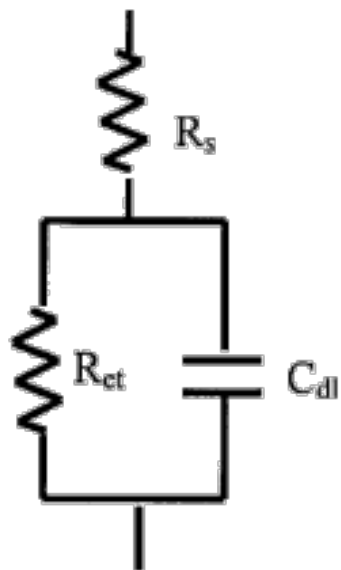
For a surface distribution:

$$C_{eff} = Q^{1/n} R_s^{(1-n)/n}$$

Hirschorn et al., *Electrochimica Acta* 55 (2010) 6218



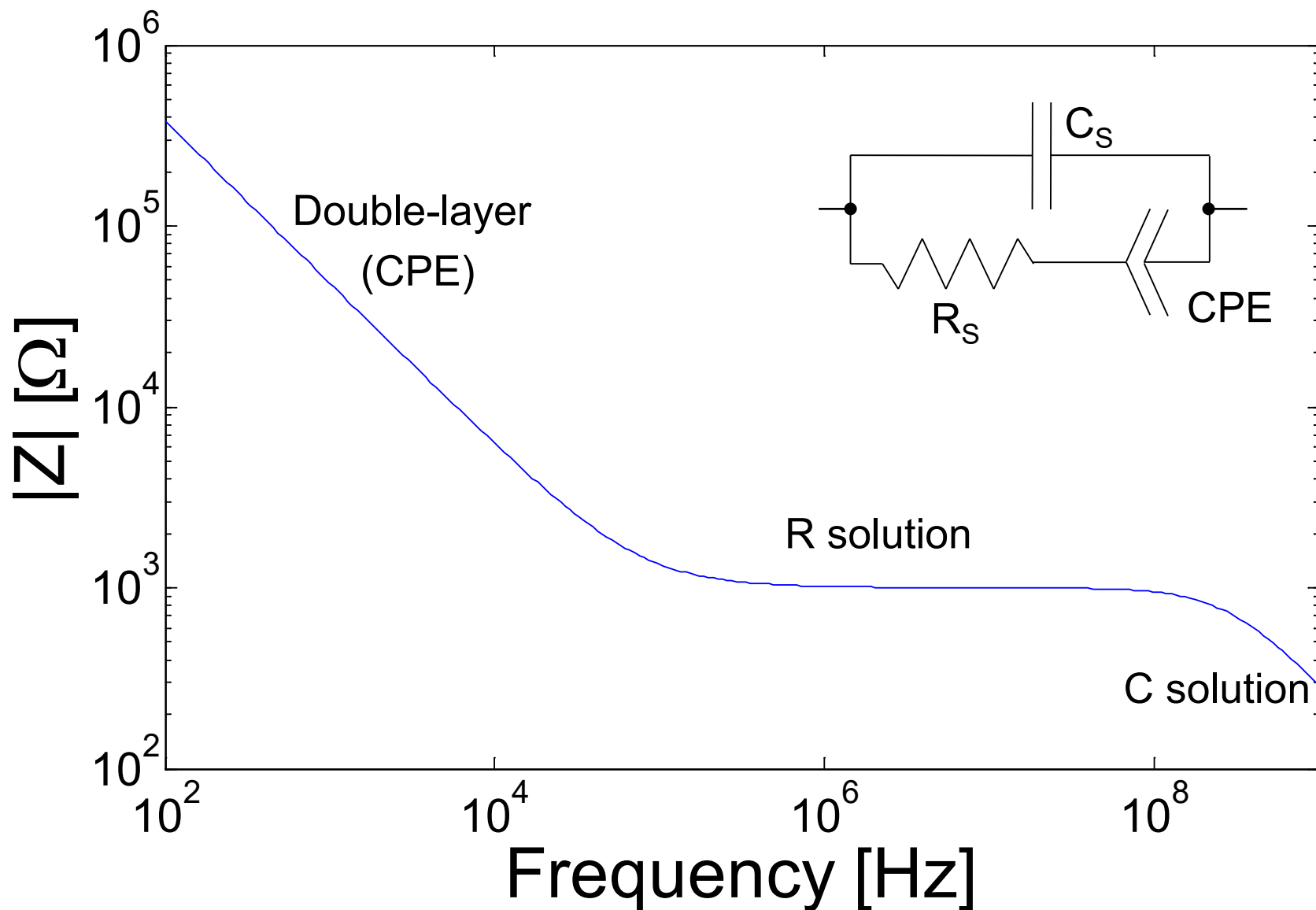
CPE: Look at the Phase



*Ding et al. / Electrochimica Acta 50
(2005) 3660–3666*



Equivalent Small Signal Model: non Faradaic

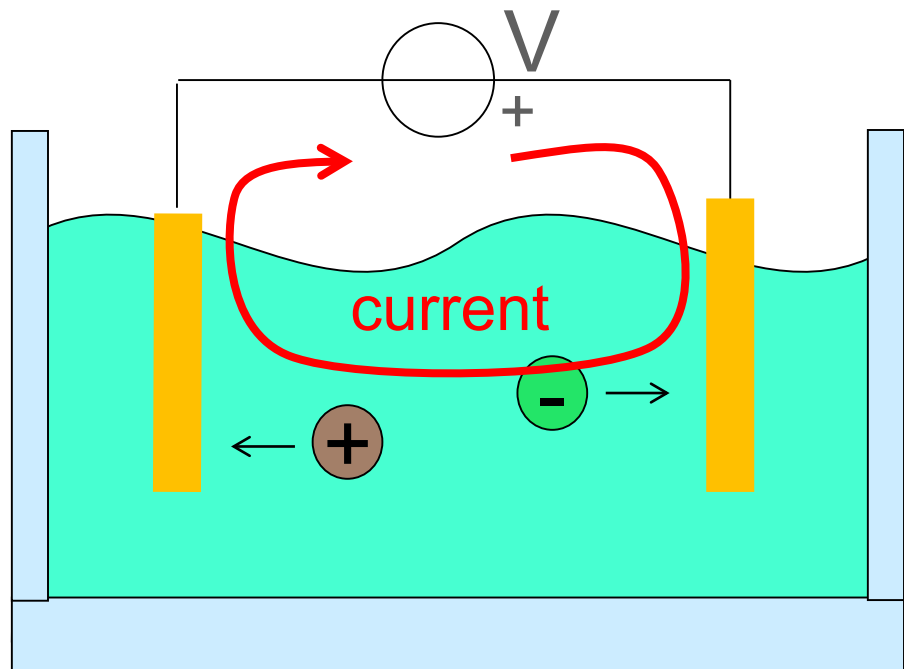




Faradaic Processes

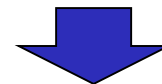


DC Electrical Current in Ionic Conductors



Ions move in the liquid

Electrons move in the electrodes



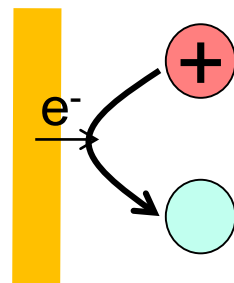
electron transfer at the metal/liquid interface

Two main cases:

«consumable» electrode:
metal grows (or thin) as a
function of time
corrosion / deposition

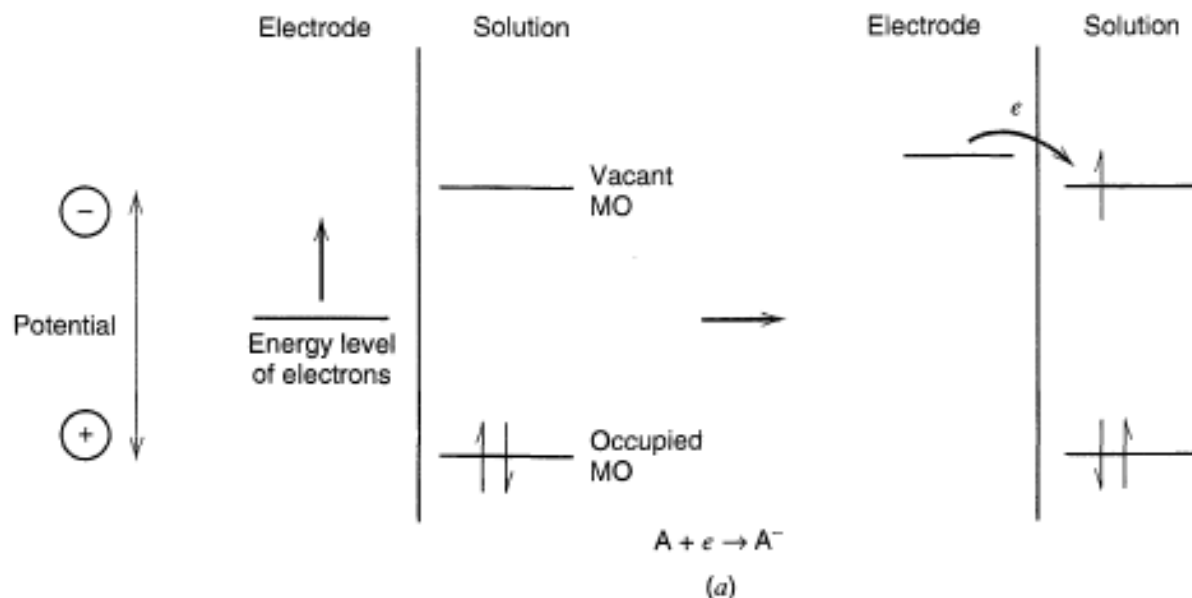


«nonconsumable» electrode: reduction-oxidation (redox) reaction of ions or molecules in liquid





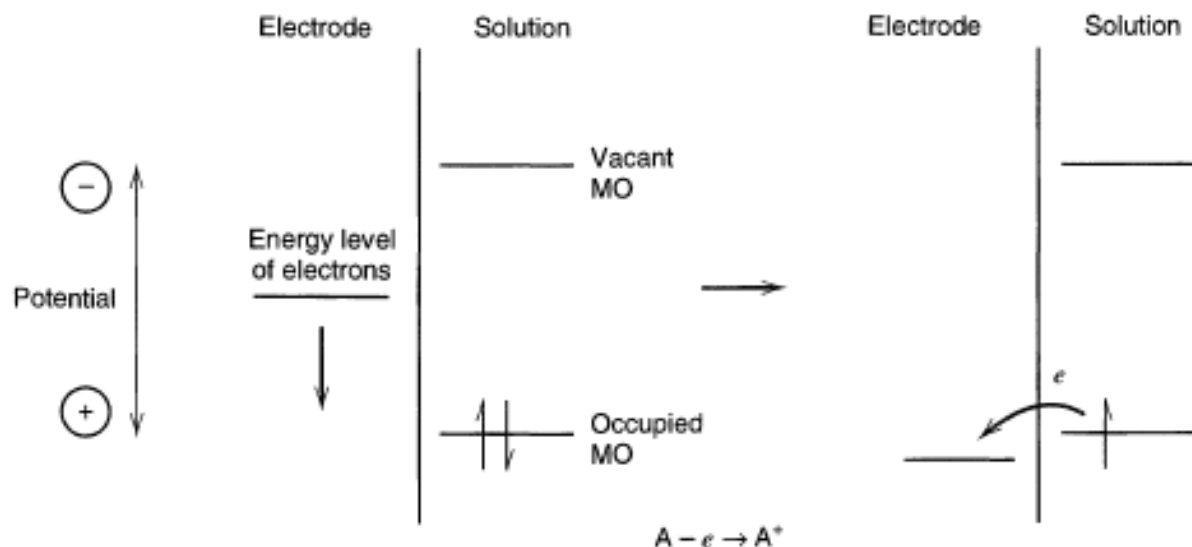
Electron Transfer



Reduction:

atom/ion/molecule
gains an electron

reaction controlled by
the potential drop
across the interface



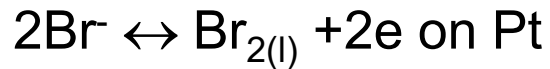
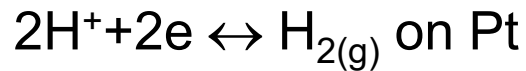
Oxidation:

atom/ion/molecule
loses an electron

Bard, Electrochemical methods.
Fundamentals and applications

Current: Mass-Transfer Effects

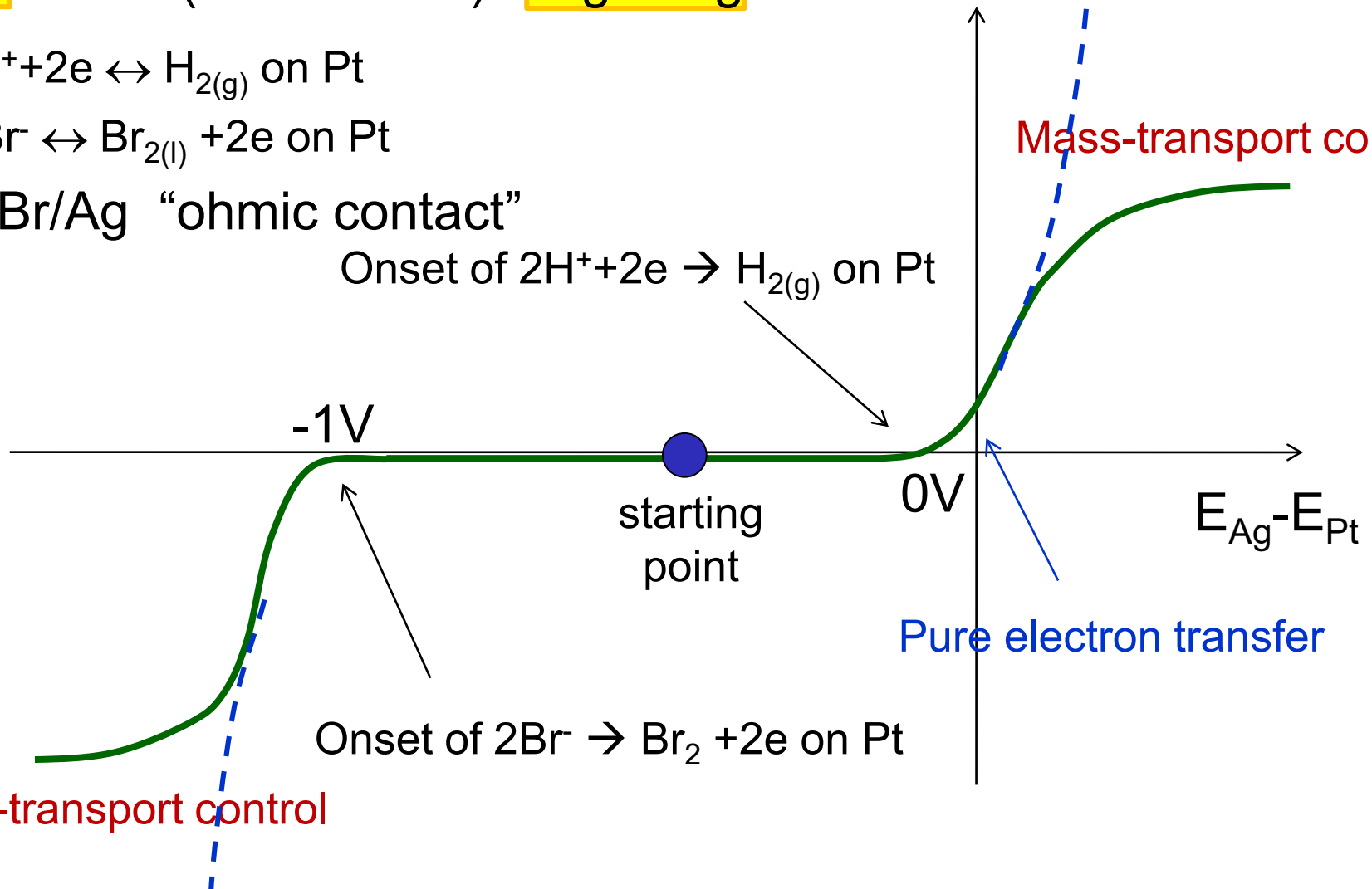
Pt - Br⁻ (solution 1M) - **AgBr/Ag**



AgBr/Ag “ohmic contact”

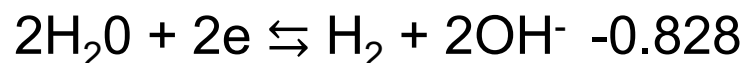
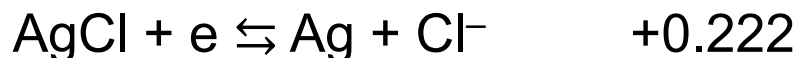
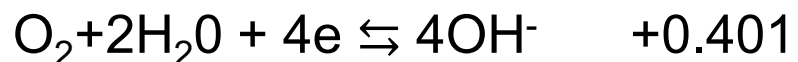
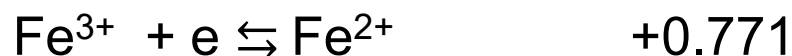
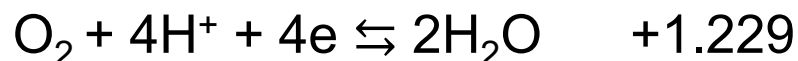
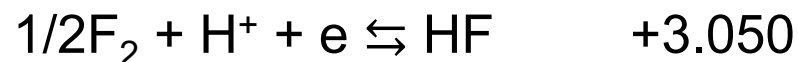
Onset of $2\text{H}^+ + 2\text{e} \rightarrow \text{H}_{2(\text{g})}$ on Pt

Mass-transport control





Voltage Ranges



Aqueous Solutions at 25°C
 $C^0=1\text{M}$

In aqueous solutions the **voltage range is limited** by the reduction /oxidation of water: $\approx 1.5\text{V}$



- Diffusion
- Migration (drift)
- Convection
 - Natural
 - Mechanical

Which is the limiting factor?

For simplicity (interpretation and setup) and/or for necessity (high ionic concentration) the mass transport of the redox species is very often controlled by **diffusion**

Response to a Potential Sweep

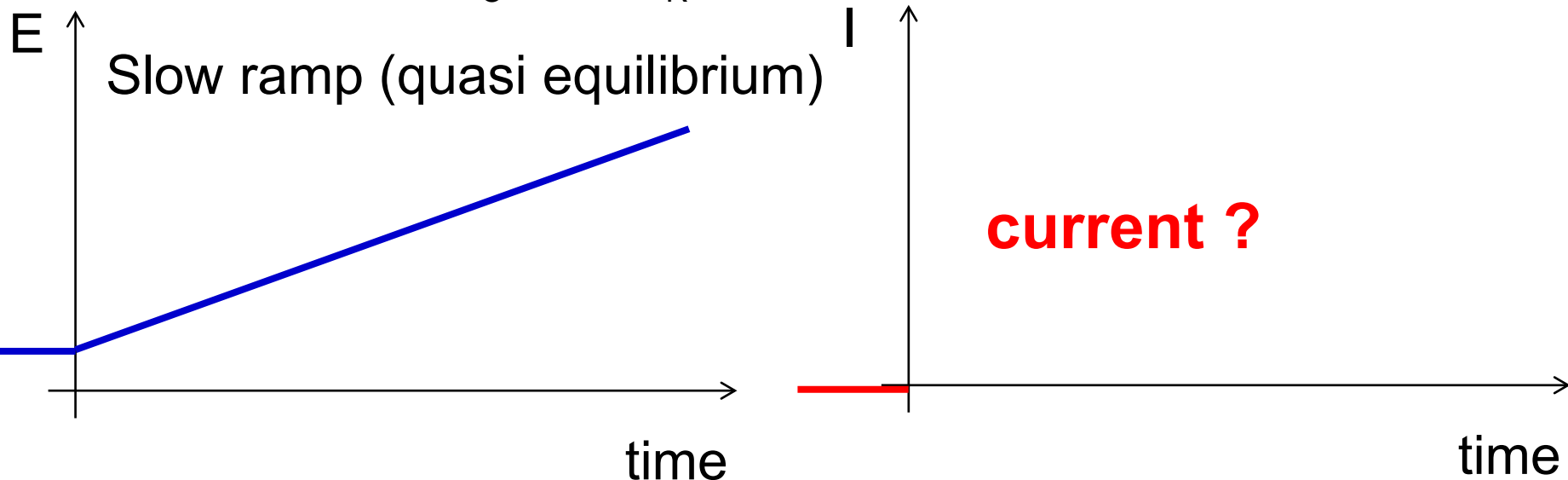
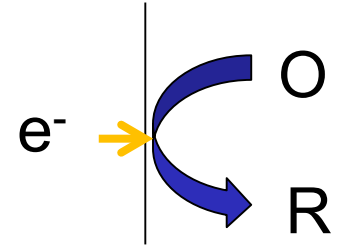
Consider: $O + e^- \rightleftharpoons R$

Assume:

$$C_R(t < 0) = 0$$

$$E(t < 0) \gg E^0$$

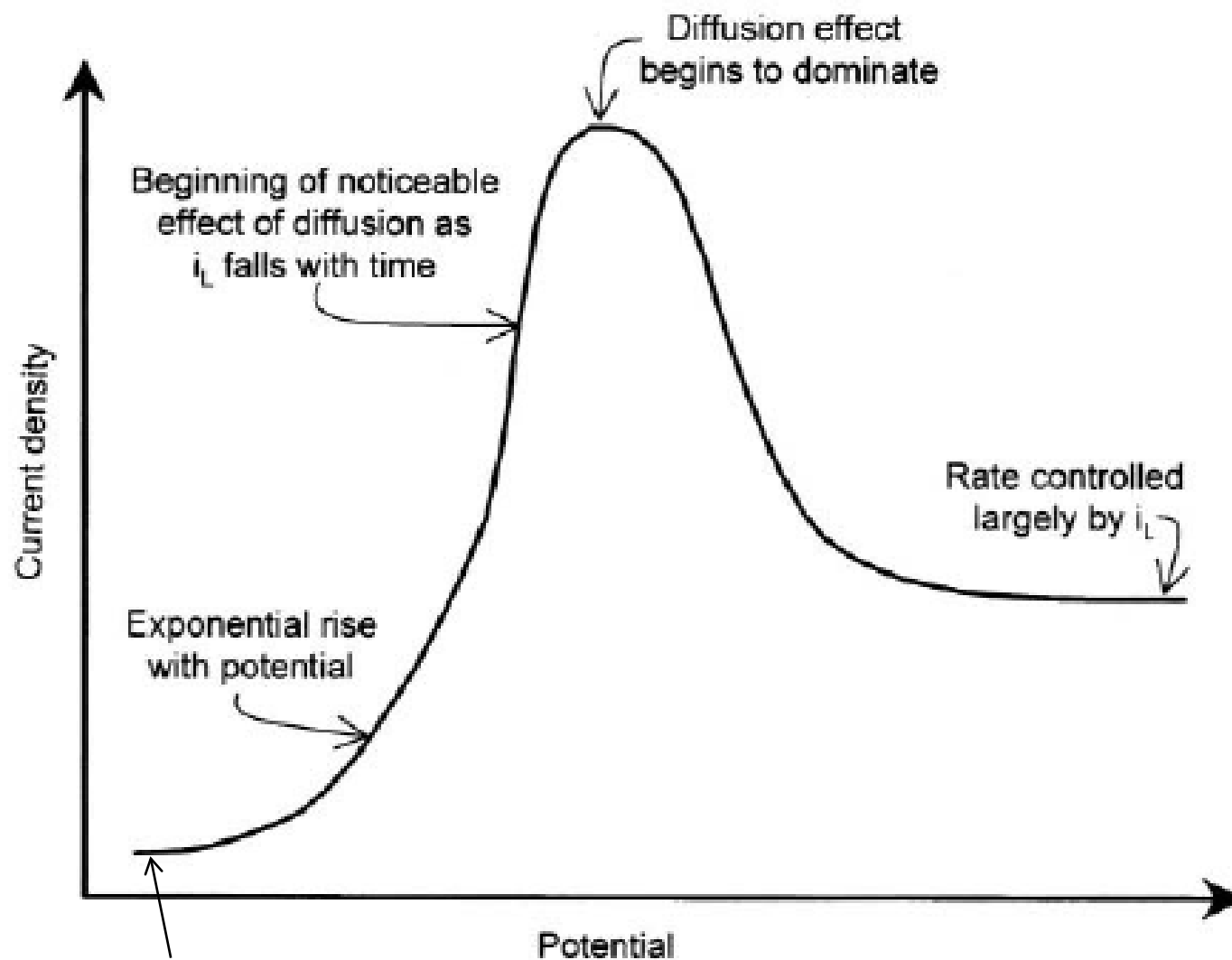
Nernstian reaction: $C_O(x=0)$, $C_R(x=0)$ determined by Nernst eq.



$$i = qN_{Av}Ak_0 \left[C_O(0,t)e^{-\alpha q(E-E^{0'})/kT} - C_R(0,t)e^{(1-\alpha)q(E-E^{0'})/kT} \right]$$



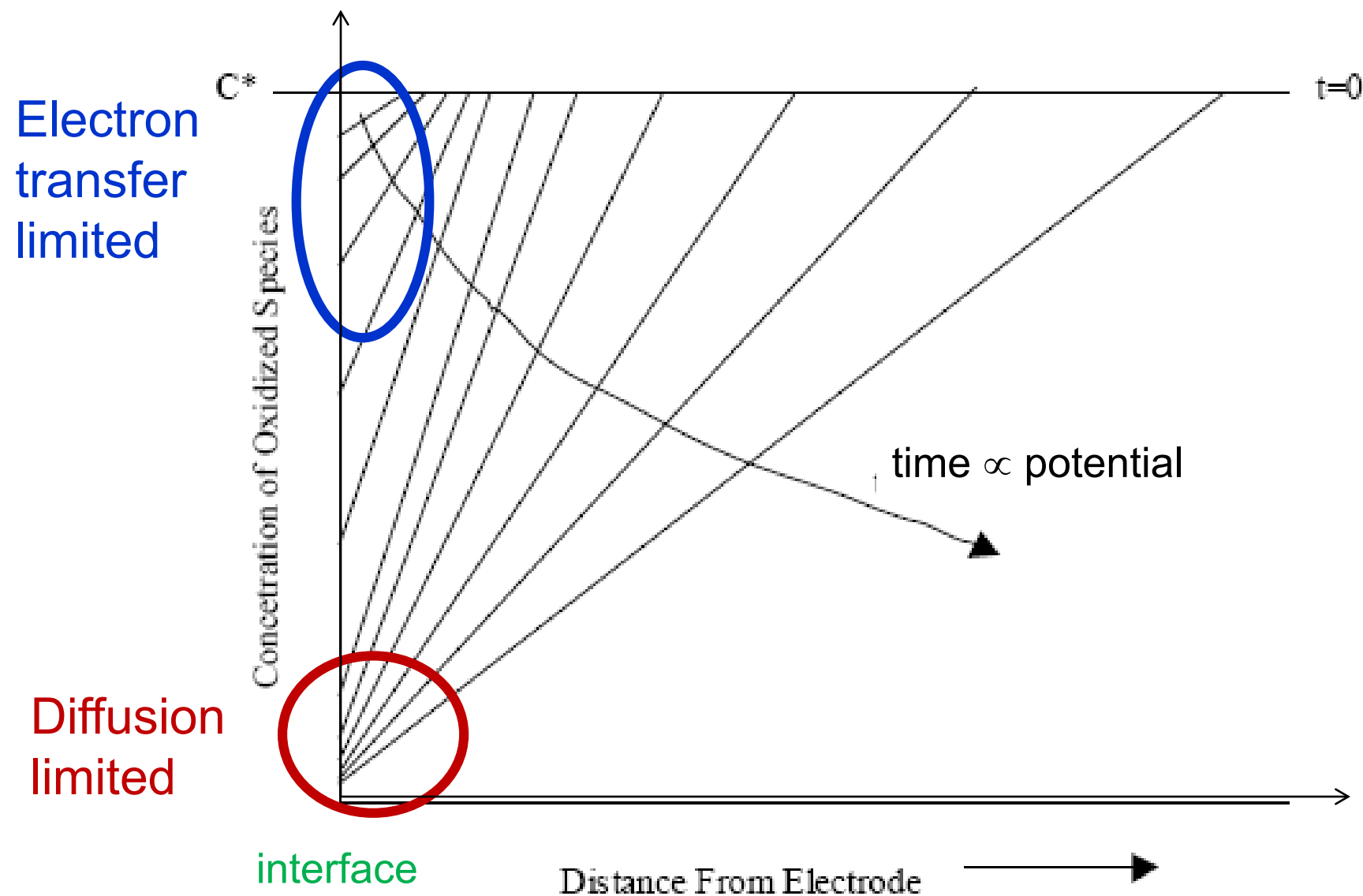
Peaked Current Profile



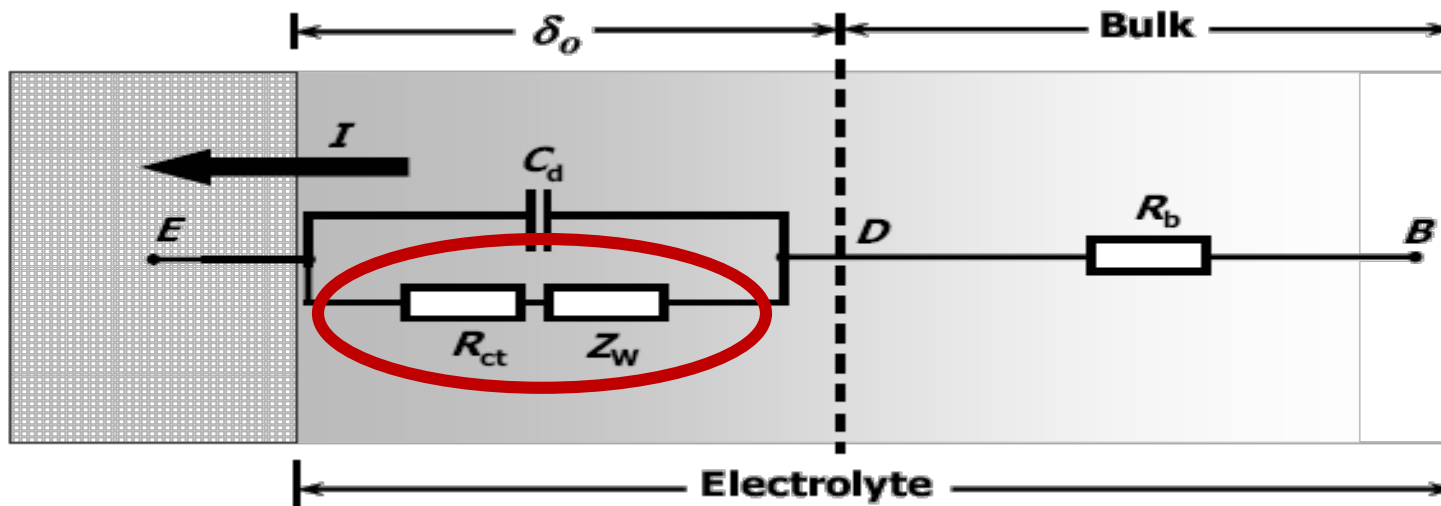
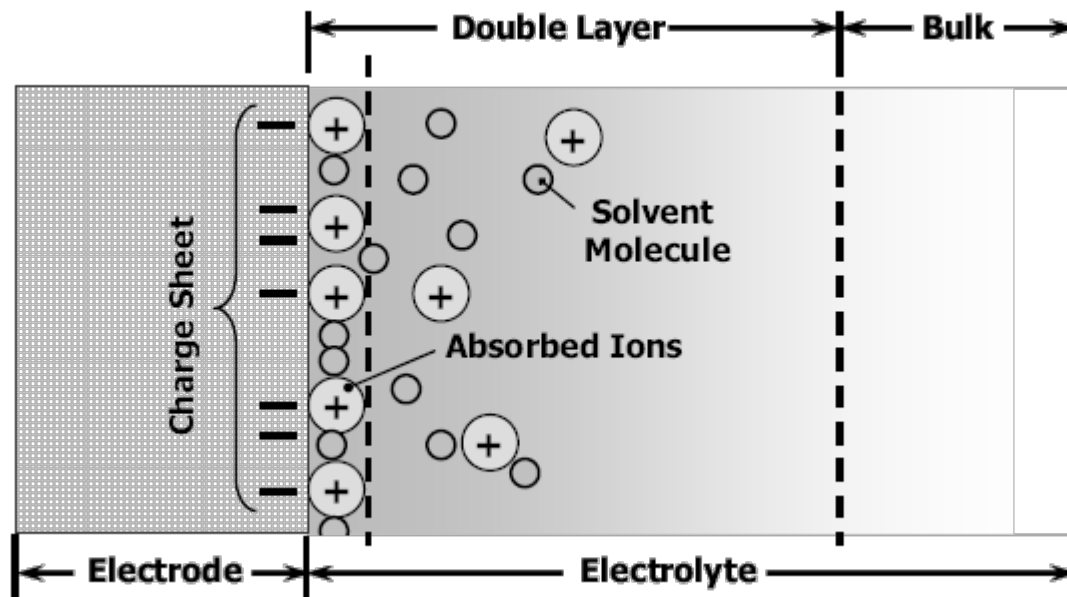
C_{dl} charging: $i = C_{dl} \cdot dV/dt$



Concentration Profile



The Interface Equivalent Circuit

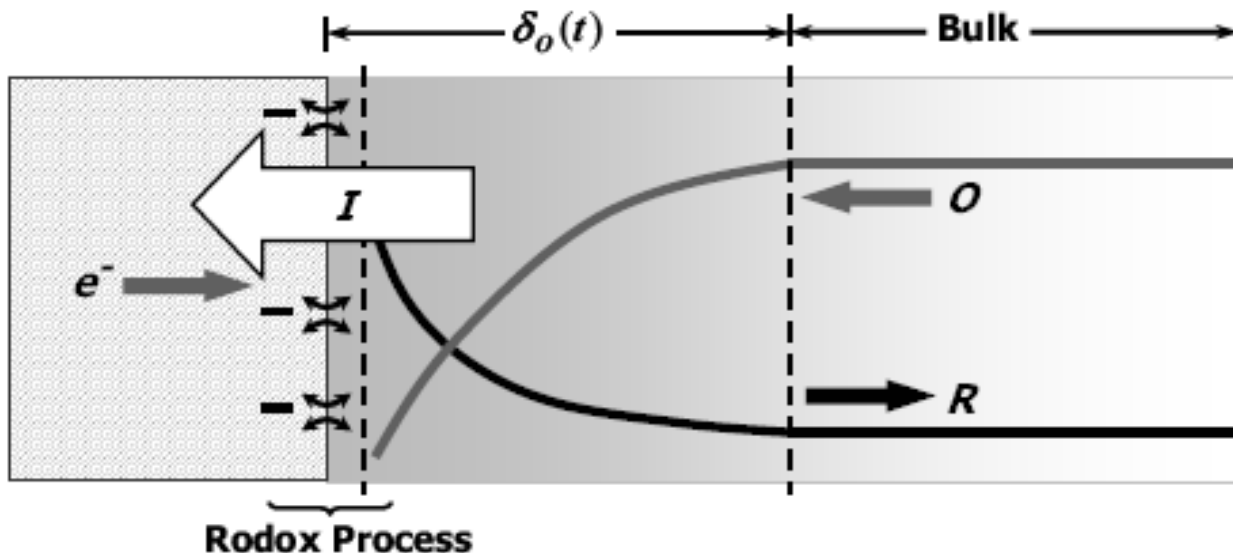


Hassibi, *J. Appl. Phys.*, 97,
084701, 2005.



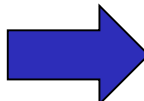
Equivalent Circuit: Charge Transfer Resistance

Charge transfer resistance requires a red-ox couple in solution:



$$i = i_0 \left(e^{-\alpha nq(E-E_{eq})/kT} - e^{-(1-\alpha)nq(E-E_{eq})/kT} \right)$$

$$i_0 = qN_{Av}Ak_0C_O^{1-\alpha}C_R^\alpha$$

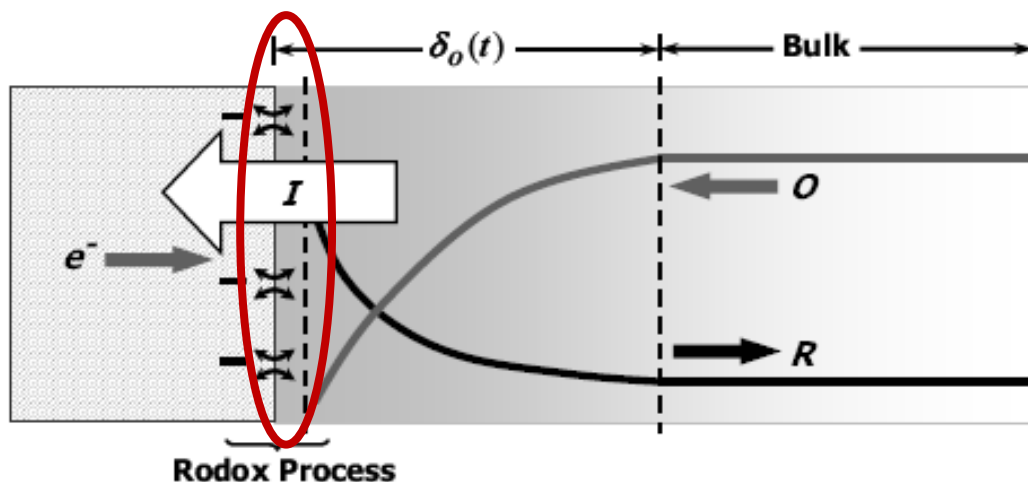
stationary condition: $E \approx E_{eq}$  $R_{ct} = \frac{\partial E}{\partial i} = \frac{kT}{nq} \frac{1}{i_0}$



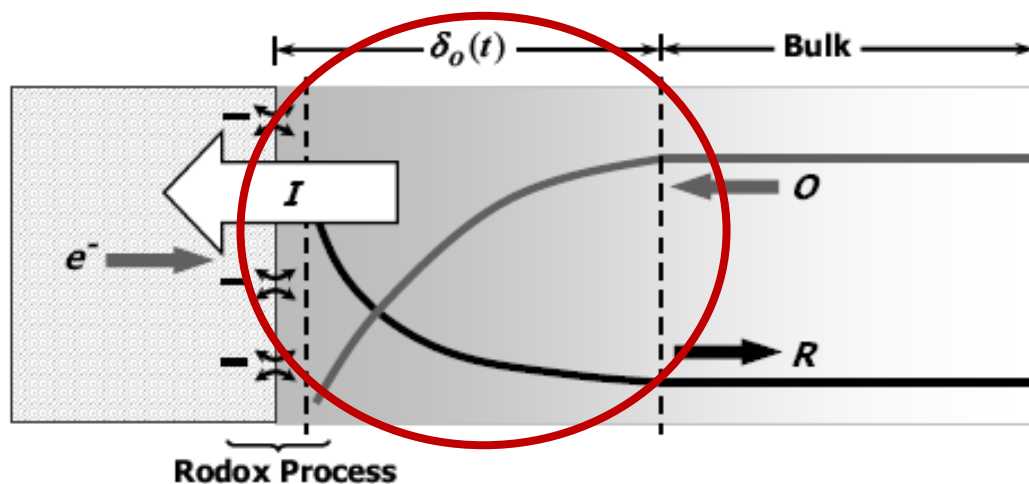
Low Frequency: Warburg Impedance

Mass-transport (diffusion) effect for long times:

“**high frequency**”: redox of the molecules at the interface
→ I limited by R_{ct}

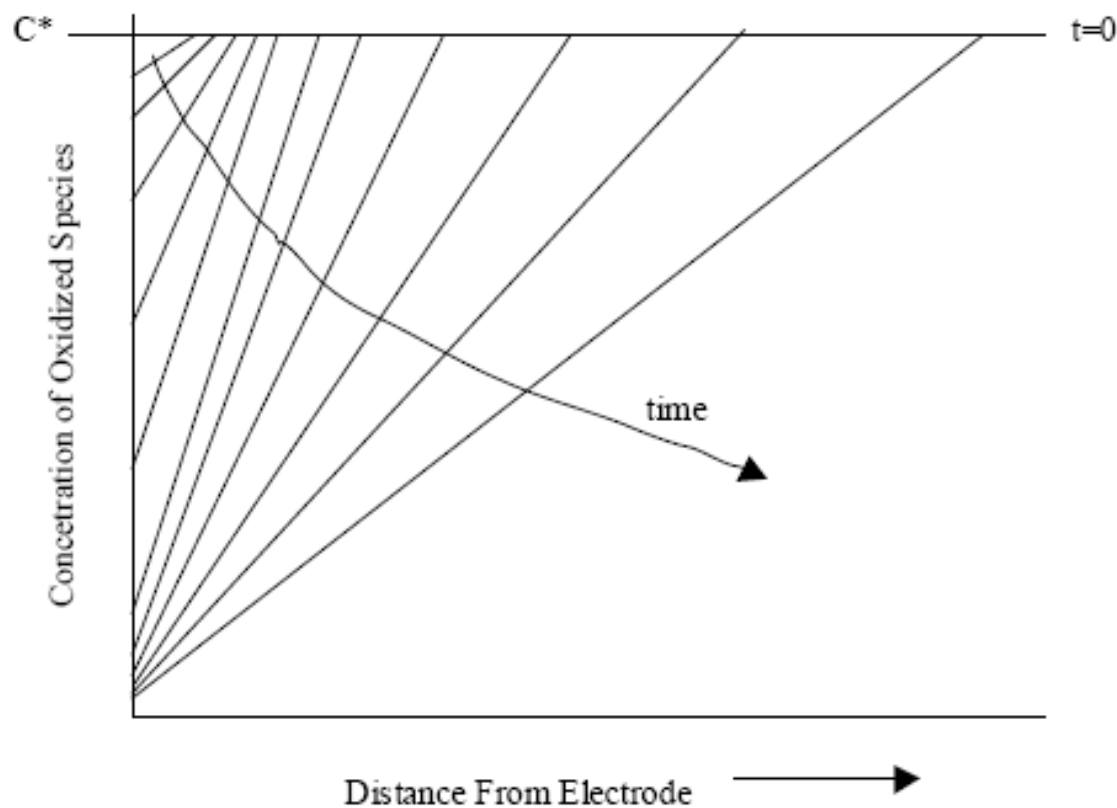


“**low frequency**”: redox requires molecules from the bulk
→ I limited by diffusion

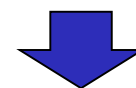




Warburg Impedance



$\delta_{\max} \approx 500\mu\text{m}$ (convection in an undisturbed solution)

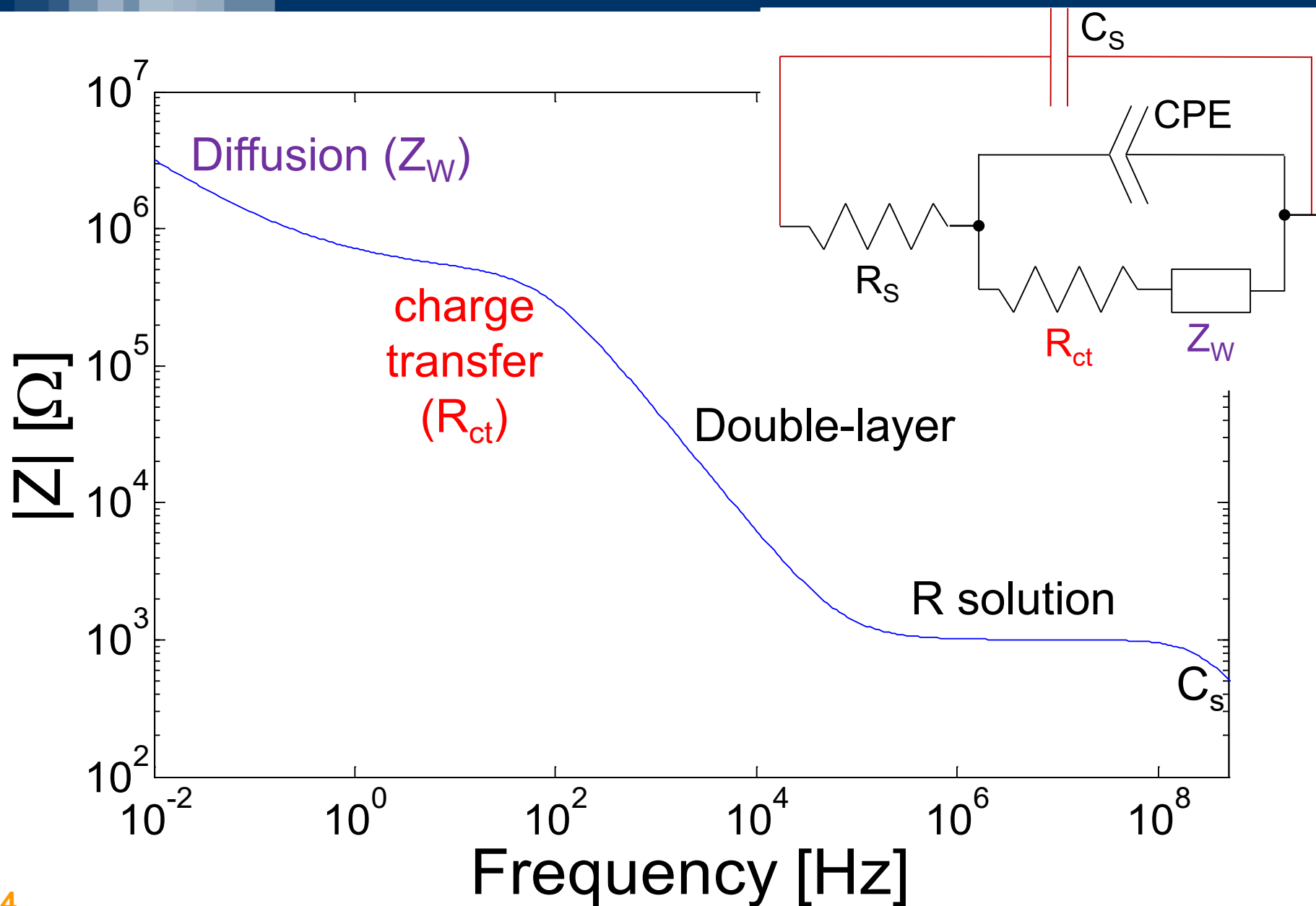


$f_{\min} \approx 1/30\text{s}$

$$\sigma = \frac{RT}{F^2 A \sqrt{2}} \left(\frac{1}{D_O^{1/2} C_O^*} + \frac{1}{D_R^{1/2} C_R^*} \right)$$



The Complete Randles Equivalent





Miniaturization



What happens when the size of the working electrode scales down to the micro (and nano) scale?

- The current to be measured is smaller (**bad**)
- The double layer capacitance is smaller (**good**)
- The mass transport limitation changes (**good**)
- The expression of the solution resistance changes ($R \sim 1/r$)
- The response time of the interface becomes faster (**good**)
- The shape of the voltammogram changes (sigmoidal)
- Single molecules can be detected!

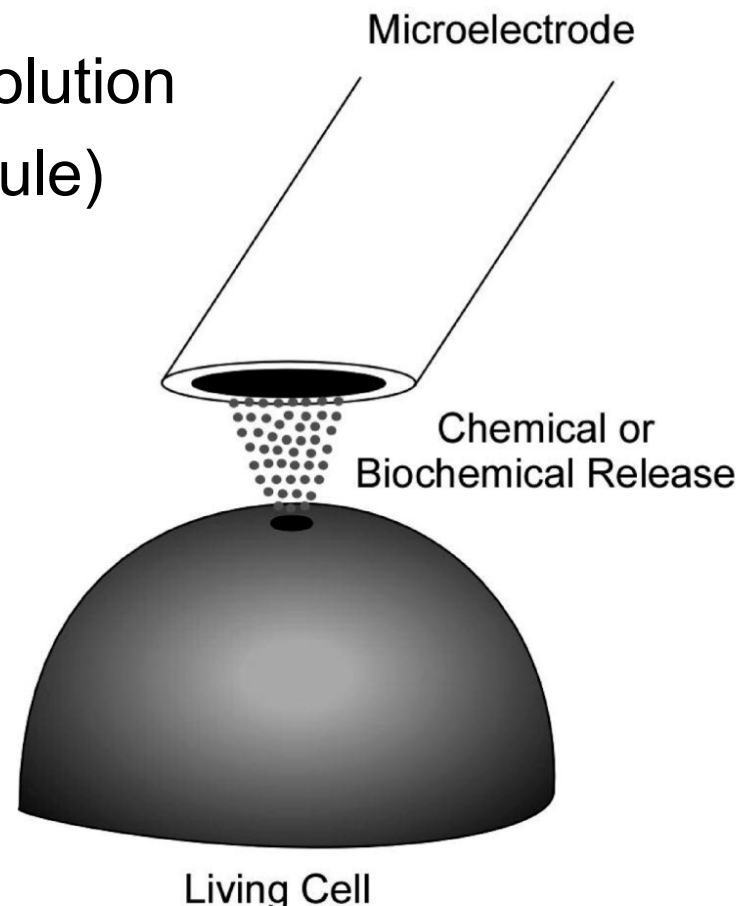
Ultra-fast voltammetry (and R_{sol} compensation) should be used

Motivation: Why Going Nano?

Micro- and nano-fabrication enables the realization of such electrodes.

Advantages:

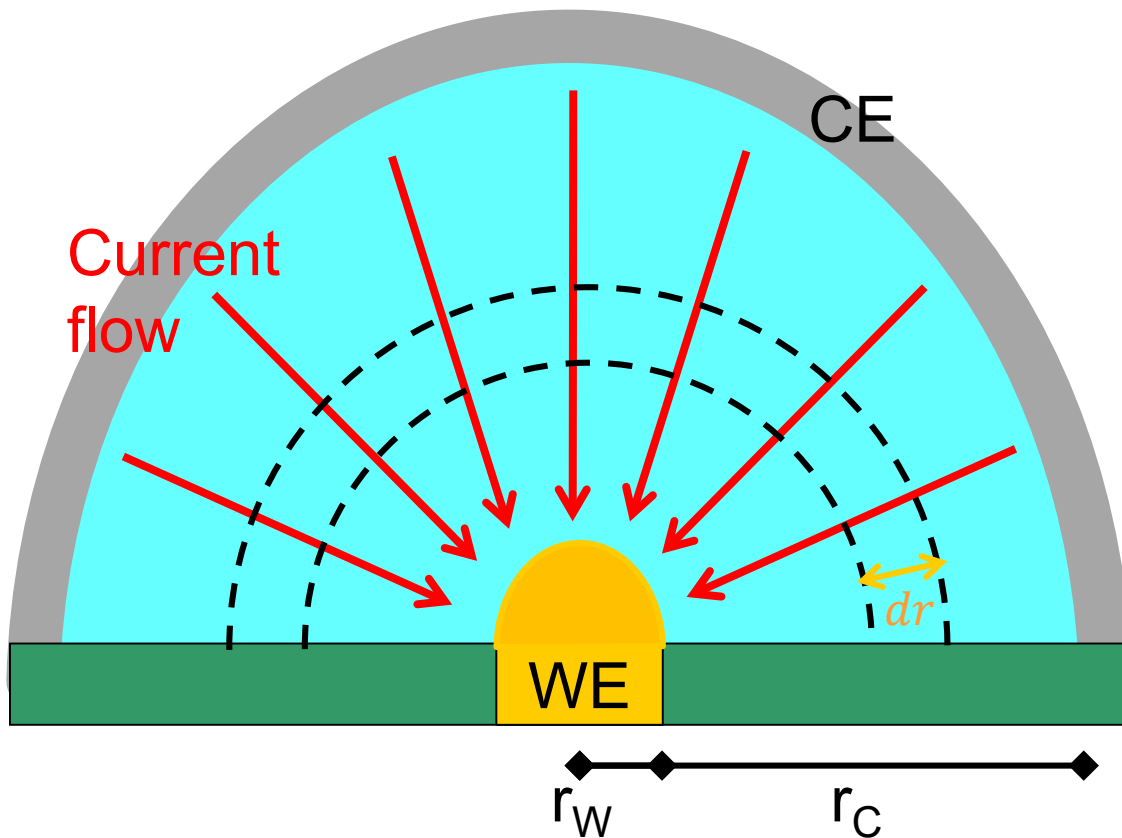
- Spatio (sub- μs)–temporal (ns) resolution
- sensitivity (down to single molecule)



Shrinking the Electrode Size

Radial diffusion profile

Hemispherical electrode:



Border effects become dominant!

$$dR(r) = \rho \frac{dr}{2\pi r^2}$$

$$R_{sol} = \int_{r_W}^{r_C} dR(r)$$

$$R_{sol} = \frac{\rho}{2\pi} \left(\frac{1}{r_W} - \frac{1}{r_C} \right)$$

↓ $r_C \gg r_W$

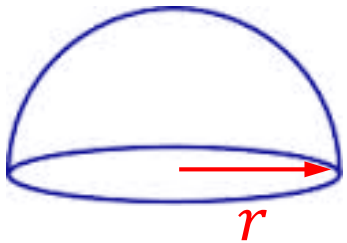
$$R_{sol} = \frac{\rho}{2\pi r_W}$$



Solution Resistance Scales with the Perimeter

$$R_{sol} = \frac{\rho}{b}$$

b = characteristic size · shape factor



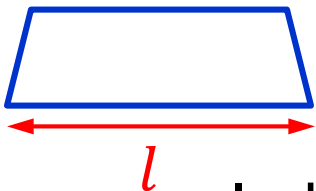
Hemispherical

$$b = 2\pi r$$



Disk

$$b = 4r$$



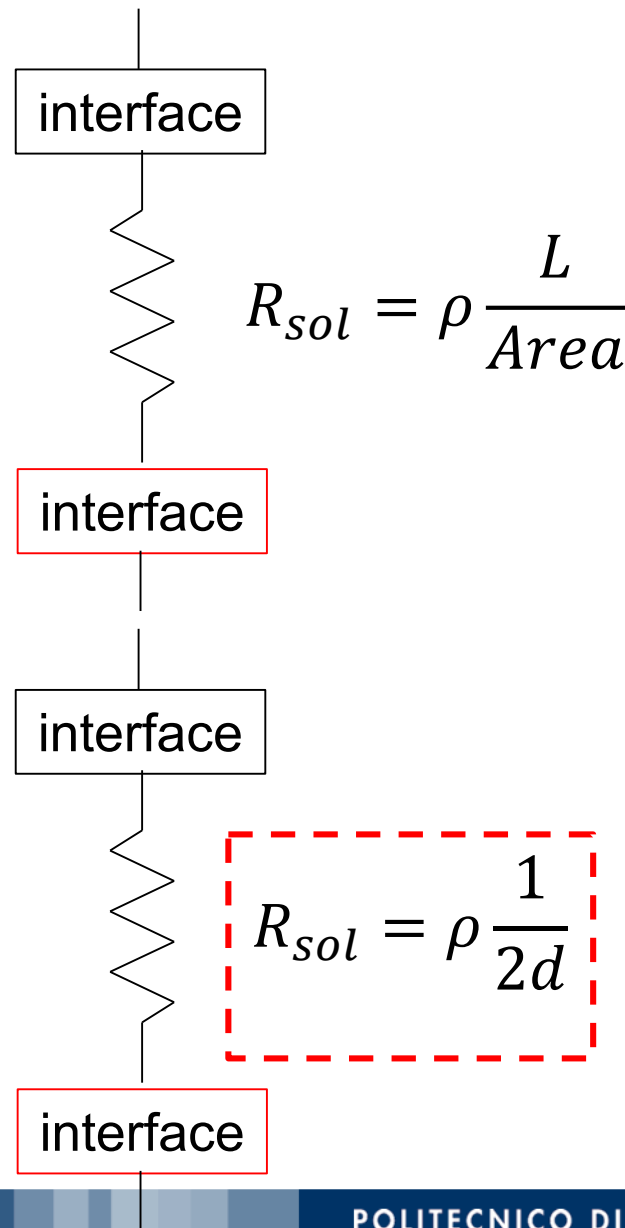
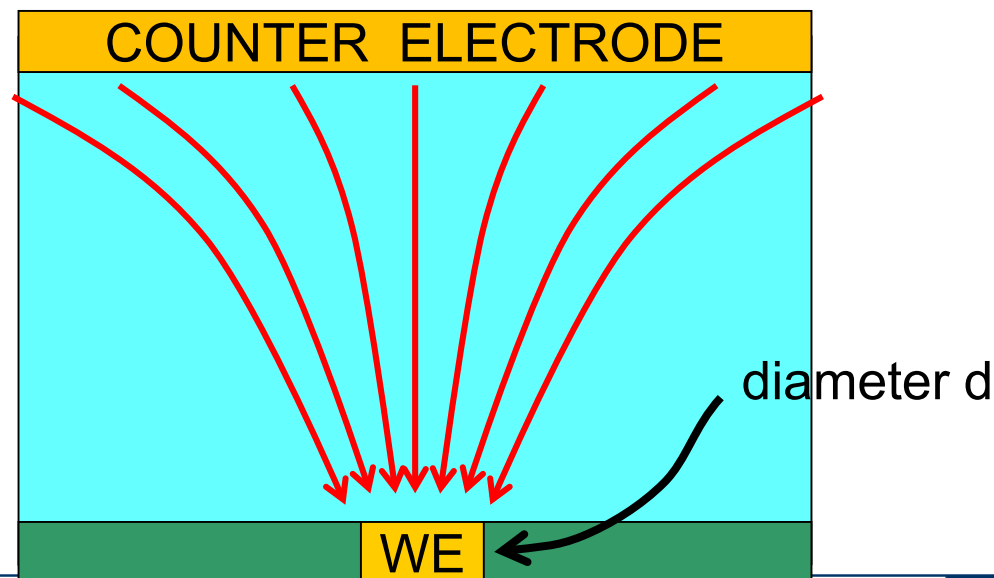
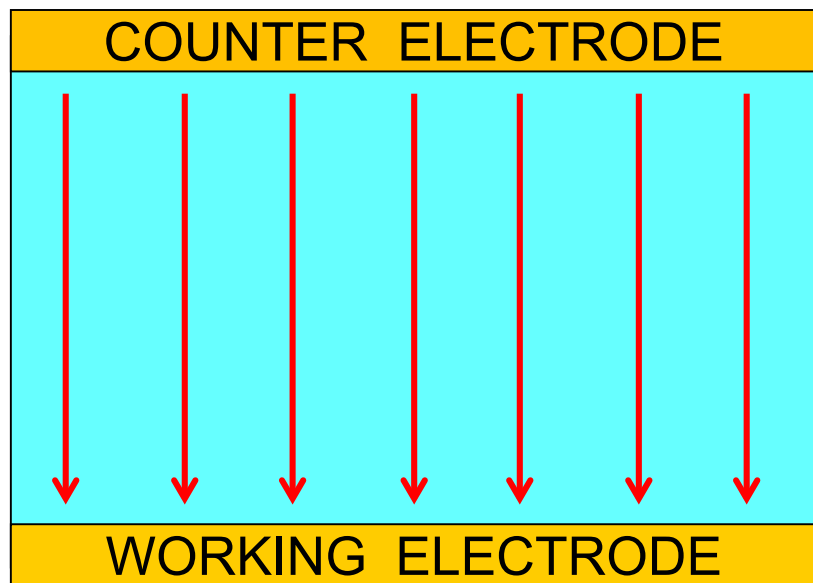
Square

$$b = \frac{\pi l}{\ln(4)}$$

Independent of the position/distance from CE



Solution Resistance for Microelectrodes





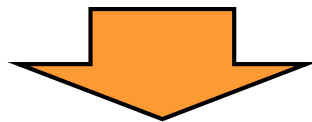
Response Time

Disk electrode (diameter d):

$$\left. \begin{aligned} C_{dl} &\cong \frac{\varepsilon}{\delta_{dl}} \cdot \pi \left(\frac{d}{2} \right)^2 \\ R_{sol} &= \frac{\rho}{2d} \end{aligned} \right\}$$

Charging time:

$$\tau = C_{dl} \cdot R_{sol} = \epsilon \rho \frac{\pi}{8} \cdot \frac{\mathbf{d}}{\delta_{dl}}$$



It becomes possible to probe faster interface phenomena

Example:	$d = 1\text{mm}$	$\rightarrow \tau = 25\mu\text{s}$
(PBS solution)	$d = 50\text{nm}$	$\rightarrow \tau = 1.2\text{ns}$